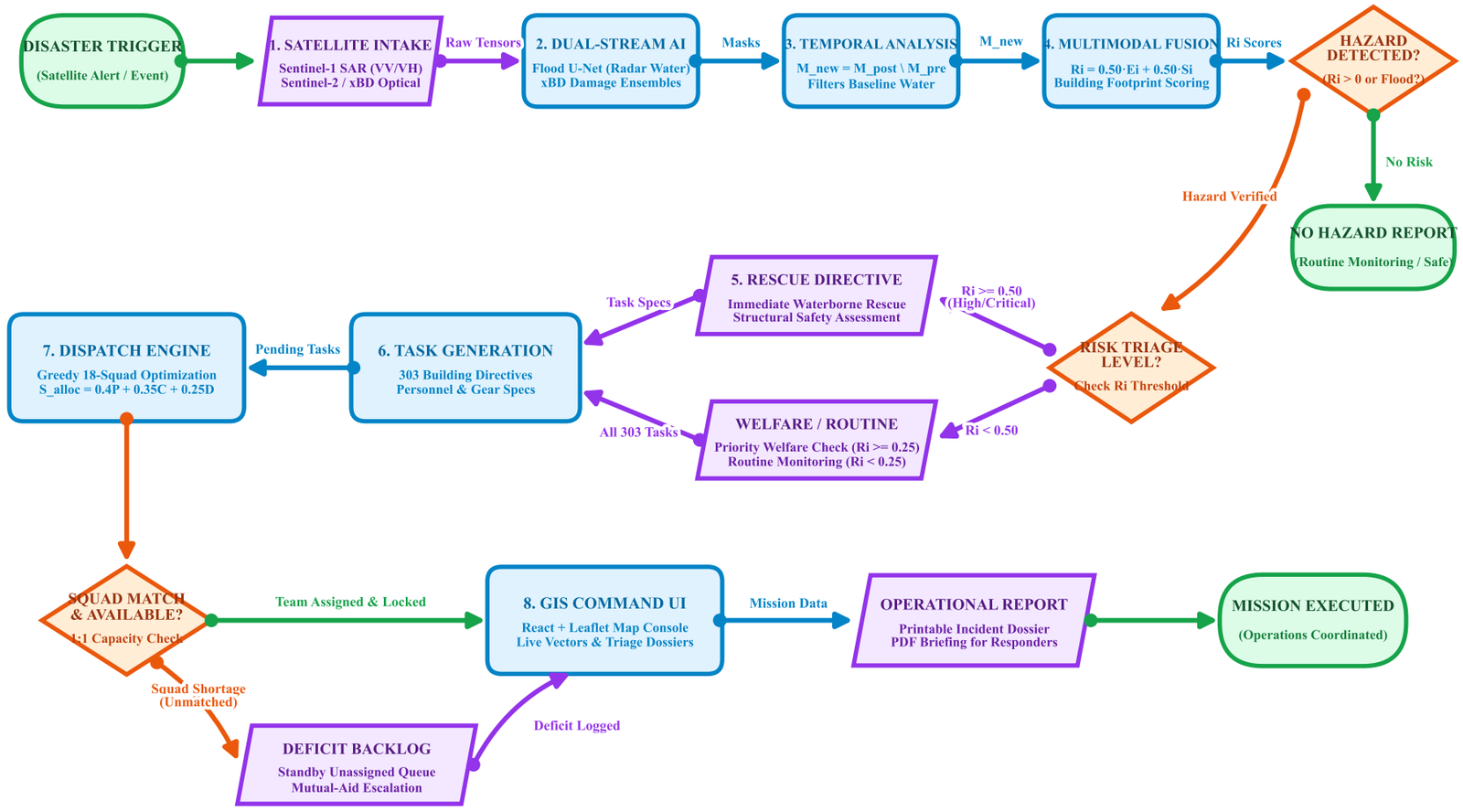
Proposed Solution

System Architecture & Operational Pipeline: Dual-Stream Remote Sensing, Multimodal Risk Triage, and Automated USAR Fleet Dispatch

A. OPERATIONAL LOGIC & SYSTEM DATA FLOWCHART



B. SEQUENTIAL IMPLEMENTATION PROCESS

1. Ingestion & Pre-Disaster Trigger — Sensor Ingest
Automated intake of Sentinel-1 SAR (VV/VH) and Sentinel-2 optical multi-band GeoTIFFs.

2. Dual-Stream AI Perception — FastAPI AI
Dual inference via Flood U-Net (radar inundation) and xBD building damage classification ensembles.

3. Temporal Baseline Analysis — Spatial Core
Differential spatial analysis M_new = M_post \ M_pre eliminates permanent water to isolate flood zones.

4. Multimodal Risk Fusion (Ri) — Risk Engine
Calculates building risk score Ri = 0.50·Ei + 0.50·Si fusing flood depth with structural destruction.

5. Risk Triage & Directive Rules — Triage Core
Threshold triage separates critical waterborne rescues (Ri >= 0.50) from routine monitoring sectors.

6. 303 Building Task Directives — Task Planner
Synthesizes 303 localized structural directives specifying required boat, crew, and gear specs.

7. 18-Squad Fleet Optimization — Dispatch Core
Greedy 18-squad allocation (S_alloc = 0.4P + 0.35C + 0.25D); unmatched cases route to deficit escalation backlog.

8. GIS Tracking & Mission Close — Rescue Teams
React Leaflet console tracks operations live, exports PDF briefing dossiers, and logs resolution.